NETWORK ROUTING AND OSPF

Assignment

Name: _____ Class: _____

Date: _____

NETWORK

TASK 1

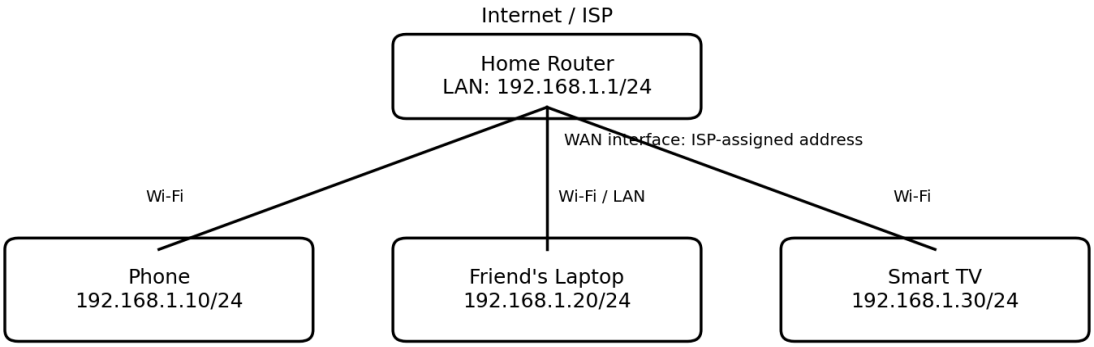
QUESTION

Draw a simple network diagram showing three devices (phone, friend's laptop, smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

ANSWER

Network diagram:

See the diagram below. It uses a sample private LAN, 192.168.1.0/24.



Example private LAN: 192.168.1.0/24
Router LAN interface: 192.168.1.1

Address labels:

Phone: 192.168.1.10/24; Friend’s laptop: 192.168.1.20/24; Smart TV: 192.168.1.30/24; Router LAN interface/default gateway: 192.168.1.1/24. The router’s WAN interface receives an address from the internet service provider, so its exact address depends on the ISP and is not shown as a fixed value.

Explanation:

All three devices are on the same local subnet and can communicate through the router’s LAN/Wi-Fi functions. The router forwards traffic destined for other networks toward the ISP through its WAN interface.

TASK 2

QUESTION

Explain step by step how a packet travels from your laptop to a friend’s computer in a different subnet. Include routers and routing-table entries checked at each hop.

ANSWER

Example network:

Your laptop: 192.168.1.10/24, default gateway 192.168.1.1. Friend’s computer: 192.168.2.20/24. Router R1 connects 192.168.1.0/24 to R2 over 10.0.12.0/30. Router R2 connects to 192.168.2.0/24.

Step 1 — Check local subnet:

The laptop compares its own subnet with 192.168.2.20. Because the destination is outside 192.168.1.0/24, it sends the packet to its default gateway, 192.168.1.1. It resolves the gateway’s local MAC address using ARP (for IPv4).

Step 2 — R1 routing decision:

R1 checks its routing table and chooses the most specific matching route for 192.168.2.0/24. Example entry: 192.168.2.0/24 via 10.0.12.2, outgoing interface toward R2. R1 forwards the packet to R2.

Step 3 — R2 routing decision:

R2 receives the packet and checks its routing table. Example entry: 192.168.2.0/24 is directly connected on its LAN interface. R2 forwards the packet onto the friend’s subnet and resolves the friend’s MAC address using ARP if needed.

Step 4 — Return traffic:

The friend’s computer sends replies to its own default gateway. The routers need a route back to 192.168.1.0/24, for example R2 route 192.168.1.0/24 via 10.0.12.1 and R1’s connected route for 192.168.1.0/24.

Sample routing-table entries:

R1: 192.168.1.0/24 — directly connected — LAN interface; 10.0.12.0/30 — directly connected — link interface; 192.168.2.0/24 — next hop 10.0.12.2 — link interface.
R2: 192.168.2.0/24 — directly connected — LAN interface; 10.0.12.0/30 — directly connected — link interface; 192.168.1.0/24 — next hop 10.0.12.1 — link interface.

Routing rule:

A router generally chooses the matching route with the longest prefix (most specific destination network). If no matching route exists, it may use a default route or drop the packet.

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TASK 3

QUESTION

Create a routing table for a fictional Zomato-style delivery network with three branches in different cities. Specify network address, subnet mask, next hop, and interface for each route.

ANSWER

Scenario:

A fictional delivery company has branches in Ahmedabad, Mumbai, and Pune, connected through a central router.

Routing table:

Routing table:

Ahmedabad LAN: 10.10.1.0/24 | Mask: 255.255.255.0 | Next hop: Directly connected | Interface: G0/0
Mumbai LAN: 10.10.2.0/24 | Mask: 255.255.255.0 | Next hop: 10.0.0.2 | Interface: G0/1
Pune LAN: 10.10.3.0/24 | Mask: 255.255.255.0 | Next hop: 10.0.0.6 | Interface: G0/2

Explanation:

The Ahmedabad subnet is directly connected to the central router. Mumbai and Pune are reached through their next-hop routers. Interface names are examples and may differ on real equipment.

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TASK 4

QUESTION

Compare static routing and dynamic routing protocols such as RIP or OSPF by listing three pros and three cons of each, using real-world service examples.

ANSWER

Static routing — Pros:

1. Simple and predictable for a small network with fixed paths.
2. No routing-protocol traffic or periodic route updates.
3. Gives the administrator direct control over the chosen route.

Example: A small restaurant branch may use a static route to reach a fixed central office server.

Static routing — Cons:

1. Routes must be changed manually when the network changes.
2. Does not automatically reroute around a failed link.
3. Becomes difficult to maintain across many branches.

Example: If a delivery company's link to a branch fails, an administrator may need to update routes manually.

Dynamic routing — Pros:

1. Routers learn routes automatically through protocols such as RIP or OSPF.
2. Can adapt routes when topology or link availability changes, depending on protocol convergence.
3. Easier to scale across many sites and subnets.

Example: A large food delivery company with many city offices can use OSPF to exchange routes between branch routers.

Dynamic routing — Cons:

1. More configuration and troubleshooting complexity.
2. Routing updates and protocol processing use network and device resources.
3. Convergence may take time, and incorrect configuration can spread bad routes.

Example: A misconfigured route advertisement could temporarily affect access to an internal ordering service.

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TASK 5

QUESTION

Use an AI tool to generate a sample OSPF configuration between two routers and explain what each configuration line does.

ANSWER

Topology:

Router R1 LAN: 192.168.1.0/24; R1–R2 link: 10.0.12.0/30; Router R2 LAN: 192.168.2.0/24. The examples use Cisco IOS-style syntax.

R1 configuration:

```
router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
```

R2 configuration:

```
router ospf 1
 router-id 2.2.2.2
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
```

Line-by-line explanation:

router ospf 1 — starts OSPF process 1 locally on the router.
router-id 1.1.1.1 / 2.2.2.2 — sets a unique identifier for each router.
network 192.168.1.0 0.0.0.255 area 0 — enables OSPF on interfaces whose IPv4 addresses match the LAN network and advertises that network in area 0.
network 10.0.12.0 0.0.0.3 area 0 — enables OSPF on the /30 inter-router link and places it in area 0.
Area 0 is the OSPF backbone area.

Verification:

On each router, use show ip ospf neighbor to check whether the neighbor relationship forms, and show ip route ospf to view OSPF-learned routes. Exact interface details and configuration modes vary by router platform. The sample is for learning and should be adapted to the actual topology.

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